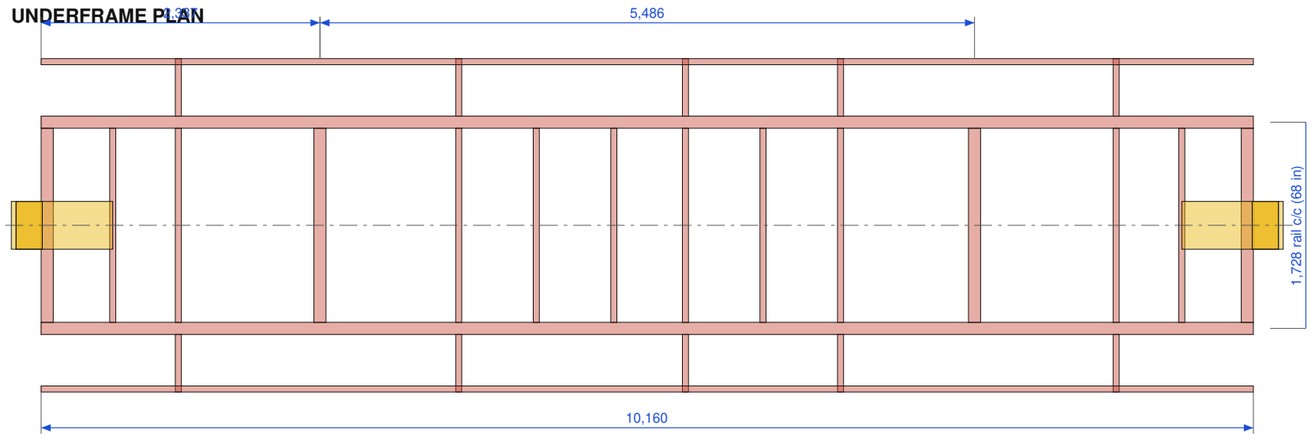
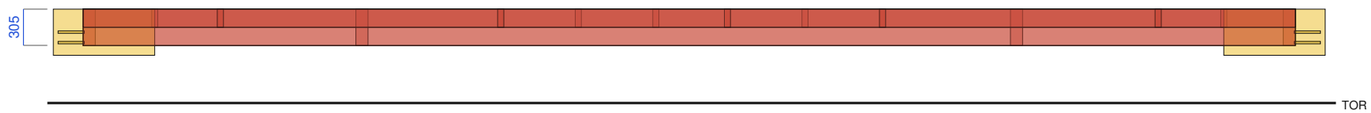
UNDERFRAME PLAN



SIDE



MEMBER SCHEDULE (matches J1a study sections)

#	Part	Section / material	Qty	Length	Location
1	Main rail	HSS12x4x1/4 A500C	2	10,160	Y±864
2	End sill	HSS12x4x1/4 A500C	2	1,626	
3	Body bolster	HSS12x4x1/4 + 2 PL10x100 + diaphragms	2	1,626	X2337 / X7823
4	Crossmember	HSS6x2x3/16 A500C	10	1,626	see plan
5	Outrigger	HSS6x2x3/16 A500C	10	432	to side sill
6	Side sill	HSS6x2x3/16 A500C	2	10,160	Y±1,371
7	PF-DB box	PL16 A572-50	2	850	pins X-100 / X10,260
8	Floor	1 in composite panel, screwed + bonded	1		

NOTES

Checks: main rail 48 MPa (×1.3), 74 MPa (×2.0); +70 kN buff 64 MPa.
Bolster 77 / 118 MPa; end deflection 1.7 mm.
Crossmembers kept clear of truck sweep (±1,000 of bolster).
Coupler CL 550 lies inside rail depth 483-787 → no drop needed
beyond the 850 box; no eccentric hitch (fix F02).
All tube ends capped; drain holes Ø10 at low points.
Jacking pads under bolster ends (4).

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Coach underframe and member schedule

Sheet C-302 · REV L2 · 2026-10-27 · 1:50 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED